

Marangduth NWFP management and marketing group



Name of NWFP Group: Marangduth NWFP Group

Village: Marangduth

Gewog: Ngangla

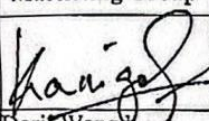
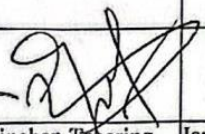
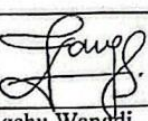
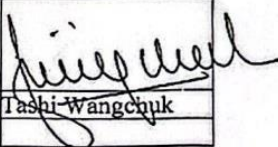
Dzongkhag: Zhemgang

Plan Period: 2024 – 2034 (*1st Revision*)

Approval Sheet

Name of NWFP Management and Marketing Group	Marangduth NWFP Management and Marketing Group
Village(s)	Marangduth
Gewog	Ngangla
Dzongkha/Dungkhag	Zhemgang
No. of households	28
Area (ha)	35
Plan period	2024-2034

Prepared and submitted for Approval

	Chairperson of NWFP Management and Marketing Group	Gup of the Gewog	Head FRPMS of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
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Date:				

Reviewed and recommended for Approval

	Reviewing Officer, FRPMD	Chief Forestry Officer, FRPMD
Signature:		
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Date:	25-7-2024	25/07/24

APPROVED BY



DIRECTOR

Department of Forests and Park Services

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Acronyms

BDBL	Bhutan Development Bank Limited
DFO	Divisional Forest Office
hh	Household
ITMS	Institute of Traditional Medicine Services
Kg	Kilogram
Km	Kilometer
ac	Acre
ha	Hectare
KNC	Khenrig Namsum Cooperatives
NWFP	Non-wood Forest Product
FRPMD	Forest Resources Planning and Management Division
FRPMS	Forest Resources Planning and Management Section

Part A: Non-Wood Forest Produce Management and Marketing Plan for Marangduth NWFP Group

1. Background

Marangduth Non-wood Forest Management (NWFP) Group is located under Ngangla Gewog, Zhemgang Dzongkhag. It is about ten minutes' drive from Ngangla Geog Center. The community has access to modern amenities like hospital, higher secondary school, RNR-Extension Centre, forestry office, electricity, water supply, motorable road and telecommunication network. Agriculture is primary source of livelihood in the village and people cultivate maize, rice, potatoes, beans, cabbage, chilly and cash crops like orange and cardamom. The community also depend on forest for timber, firewood and NWFPs for additional income generation for their livelihood. The village has 28 households and all are registered member of Marangduth NWFP Management and Marketing Group. There are various NWFPs found in their locality such as *Neomicrocalamus andropogonifolius* (Yula), *Diplazium esculentum* (Jeema), *Knema tenuinervia* (Durka), *Borinda grossa* (Baa), *Piper betleoides* (Dhem pan), *Oroxylum indicum* (Namkaling), *Thysanolaenna maxima* (Tsakusha), *Musa and its parts* (Mochali and pachali), *Tupistra nutans* (Ngawa pem/Thorma Ju), *Cane species* (Khran khengkha) and *Dendrocalamus hamiltonii*. Though community harvest all these species for self-consumption, and few for commercial purposes. They prefer and prioritized *Dendrocalamus hamiltonii* for extensive commercialization as the group is equipped with facilities for bamboo processing. There are four bamboo management areas consisting 35 ha. which will be management by community during the plan period.

During the first management plan period, the community couldn't conduct the activities as per plan due to Covid-19 pandemic. However, the group has harvested and sold the bamboo shoot. The previous plan had expired in June, 2022. Before reviewing the management plan, a comprehensive consultation meeting with the management group members was conducted to understand the inactiveness and to assess their interest in continuing further managing the NWFP. Capturing their interest in continued management of NWFPs, Divisional Forest Office (DFO). Zhemgang provided technical support to the group in resource assessment and preparation of management plan. The management plan will guide the group members to implement the activities to ensure the proper harvesting technique and other management practices.

2. Goal and objectives

In addition to the objectives reflected in the LFMA, the cross-cutting management plan shall be managed as per the following objectives:

- To uplift the rural economy of the community through sale of NWFPs resources.
- To utilize the resources on sustainable basis.

3. Description of NWFPs present in the area demarcated for management

Dendrocalamus hamiltonii: It is one of the bamboo species with straight strong (near solid) culms and edible shoots. It has tight clumping growth habit and slightly weeping tops with lush foliage. Some of the culms may have a cream striation on the lower inter-nodes. Culms is 9-25cm in diameter, strong pendulous above, densely covered in persistent, brown and white, furry and wax; wall thin; branches fewer toward base, center branch to 5cm in diameter, smallest branches curving from clump, nodes with dense, long aerial roots. The group will collect the resources from Dangling, Dangling 1, Nimroe and Chackchawa blocks. It is around 30minutes to 1 hour drive from the village.

Neomicrocalamus andropogonifolius: It is a perennial plant with the following characteristics: pluricaespitose, elongated rhizomes, scandent culms, it grows up to 1200cm long, woody, culm-internodes terete, thin-walled, culm nodes flush with internodes, leaf-blade base broadly rounded, leaf-blade margins scabrous and leaf-blade apex acuminate. It is mainly used for crafting purposes. Management group wanted to supply the resources to Bjoka NWFP

management group for bamboo crafting & also wanted to train themselves for crafting to bamboo production. Due to flowering season of Yula, assessment couldn't conduct.

Diplazium esculentum: it is straggling fern that grows up to a meter tall. The young shoots or fronds of the fern are consumed as vegetables, sold in local markets throughout Asia and the Pacific Islands. Fronds are variable in size, bipinnate, linear leaflets with lobed to serrated margins. Is the vegetable fern, edible fern found throughout Asia and Oceania. It is probably the most commonly consumed fern. The management group would collect the resources within their managed area and wanted to sell fresh or processed fern in the market.

Knema tenuinervia: Is an evergreen shrub or tree, usually growing from 3 - 20 meters tall with some specimens reported to reach 30 meters. The tree is harvested for its wood, which is used locally and especially for processing into traditional medicines.

Borinda grossa: Is an evergreen Bamboo growing to 10 m (32ft) by 3 m (9ft). The flowers are hermaphrodite (have both male and female organs) and are pollinated by Wind. Suitable for: light (sandy), medium (loamy) and heavy (clay) soils and prefers well-drained soil. Suitable pH: acid, neutral and basic (alkaline) soils. It can grow in semi-shade (light woodland). It prefers moist soil.

Piper betleoides: The betel plant is an evergreen perennial, with glossy heart-shape leaves and white catkin. The betel leaf plant is a creeper grown in our country too and belongs to pepper family. Its waxy green, heart-shaped leaves are used for medicinal and culinary purposes. Community prefers the resources to collect from the vicinity of managed area for marketing.

Oroxylum indicum: Is a monotypic species of flowering plant belonging to the genus Oroxylum and the family Bignoniaceae, commonly called midnight horror, oroxylum, or Indian trumpet flower. It is a mesocaul tree which can reach a height of 18 meters. The resources are also presence within the managed NWFP area of the group.

Thysanolaenna maxima: A strongly tufted, very robust perennial grass with erect or slightly spreading solid bamboo-like culms up to 3.5 m tall. Leaf-sheath hairy along outer margin; ligule a scarious membrane, 1-2 mm long; leaf-blade lanceolate-acuminate, 30-65 cm x 3-7.5 cm, base broad and rounded or subcordate, margins scaberulous, conspicuously glaucous beneath. Inflorescence a terminal huge and drooping panicle, up to 140 cm long, well exerted, branches divided and subdivided into many branchlets; spikelets awnless, short pedicelled, falling with part of the pedicel, often in pairs on a common peduncle, 2-flowered; lower glumes clasping. Caryopsis subglobose to ovoid, 0.6 mm long, reddish-brown. Members wanted to collect the resources markets after processing.

Tupistra nutans is an amazing Indian native that makes a nice clump resembling an aspidistra on steroids. Each clump is composed of 3' long narrow dark green leaves that are held upright. The flower of species is used as vegetable which is sold expensively in the market.

Plectocomia himalayana: Stems 3-15m or more, scandent. Leaf sheaths tomentose (orange-brown in lower part, pale fawn above), heavily armed with oblique or wavy-horizontal lamellae bearing unequal (longest to 2.5cm), slender, straight spines; mouth oblique, auricles not developed. Petiole very short, not kneed. Rachis flattened; fawn-tomentose, lower part sometimes spiny, upper part developed into long cirrus (to 2m) heavily armed with grapnels. Leaves to 3.5m, pinnae relatively few (to 25), linear-lanceolate, gradually drawn into long (to 8cm) filiform apex, longest 30-45cm, widest 2-3.6cm, irregularly inserted-singly or in groups of 2-3, margins, spinulose, with many, glabrous, parallel veins.

Table 1.Brief description of NWFPs for commercial harvesting in each season

NWFP name	Harvesting area(s) (name)	Parts harvested	Use/purpose (traditional & others)	Harvesting season (months)	Estimated quantity for harvest in a season/year
<i>Neomicrocalamus andropogonifolius</i>	Demphu	Culms	Crafting	December-May	1500 bundle

<i>Dendrocalamus hamiltonii</i>	Danling, Dangling 1, Nimroe & Chackchawa	Shoot	Vegetables	May-September	3500kg
		Culm	Timber	12 months	5000 pieces
<i>Diplazium esculentum</i> (fern)	Nabang, Rangkolong, Muchari & Prikchang	Young shoot	Vegetables	March to September	1500 bundles
<i>Knema tenuinervia</i> (Durka)	Marangduth area	Woods	For religious	Whole year	3000kgs
<i>Borinda grossa</i>	Nabang Pungling Rangkolong	Culms	Mats & other products	December to February (not in shooting period)	4500pieces
<i>Piper betleoids</i>	Rangkolong	Leaf	Stimulant	Throughout the years	2500 bundles
<i>Oroxylum indicum</i>	Marangduth area Dudiphu Muchari Prikchang	Flower	Vegetables/dyestuff /Religious	December to March	2000pieces
<i>Thysonolaenna latifolia</i>	Dudiphu Muchari Dongdor Nimroy Lhadung	Upper parts, inflorescence	Fodder/soft broom	December to March	2500 bundles
<i>Musa</i>	Lhadugang Tantaling Muchari Prikchang Naykhar	Stem/Flower	Fodder/vegetables	Whole year	2500 pieces
<i>Tupistra nutans</i>	Rangkolong Reolangang Dudiphu	Flower	Vegetables	Sept-Dec	1000kg
<i>Plectocomia himalayana</i>	Ngawang Kerong Kemathung Nawgren Norbuling Teliphu	Shoot/Stem	Crafting	Dec-May	2500 pieces

4. Procedures for sustainable harvesting of NWFPs

The NWFP management group members have to adhere to specific procedure for the sustainable management of resources.

- The Chairperson will consult with forest office and obtain permit for collection during the onset of harvesting season.
- The Chairperson will announce starting date for harvesting only after obtaining required permits from forest office.
- Only one member from each household will be entitled for collection of NWFP products.
- Collection will be done within the demarcated area only.
- Harvesting prescriptions will be strictly followed by the collectors (Table 2)
- The secretary will maintain proper record of harvest and sale of NWFPs for monitoring and evaluation.

- g. Chairperson will notify the forest office on completion of harvesting the NWFPs.
- h. Treasurer of NWFPs management group to maintain all financial records.

Table 2.Brief description of NWFPs for commercial harvesting in each season

<i>NWFP Species</i>	<i>Harvesting prescription</i>
<i>Neomicrocalamus andropogonifolius</i>	▪ Cut only feasible culm ▪ Do not cut all culm & retain at least two Harvest culm when it produces secondary shoot
<i>Dendrocalamus himaltonii</i>	• Collect shoot up to 25% from the clump • Use sharp knife to cut the shoot without damaging other parts in the stand • Cover the cut portion of the shoot that is left in the stand with soil
<i>Diplazium esculentum</i> (fern)	❖ Collect the fern that is above the ground level only ❖ Do not up root the whole plants ❖ Just break the base of the fern to collect the shoot
<i>Knema tenuinervia</i>	✓ Collect only fallen trees ✓ Never cut standing trees
<i>Borinda grossa</i>	➤ Collect matured culm ➤ Do not disturb the young shoot ➤ Harvest the culm from the base to maintain sanitation
<i>Piper betleoides</i>	❖ Collect green leave ❖ Do not cut down the trees ❖ Do not break down the stem ❖ Climb up the trees to collect the leaves seeing the possibilities
<i>Oroxylum indicum</i>	• Collect flower when matured • Never cut down the trees • Handle with care while harvesting the flower
<i>Thysonolaenna latifolia</i>	➤ Collect the matured parts/stem ➤ Never cut it from the base ➤ Do not uproot the stem
<i>Musa</i>	○ Collect just matured flower ○ Do not cut down the immature stem ○ Harvest sustainably, it is easier to vanish
<i>Tupistra nutans</i>	❖ Just collect flower only ❖ Never uproot the stem
<i>Plectocomia himalayana</i>	➤ Cut only matured cane ➤ Cut canes with only 10m usable length ➤ Collect young cane shoot by bending Do not cut the tree that support the cane

4.1 Block-wise area & Distances for prioritized species

There are various collection areas for different NWFPs that will manage by Marangduth NWFPs management group but very minimal. The areas and walking distances for prioritized species especially for *Dendrocalamus hamiltonii* is as tabulated below:

Village	Collection area	Distance from village		Total area(ha)
Marangduth		Dangling	1:30hours	10
		Dangling 1	1:30hours	12
		Nimroes	30 minutes	6
		Chackchawa	20 minutes	7
Total				35ha (86.49acres)

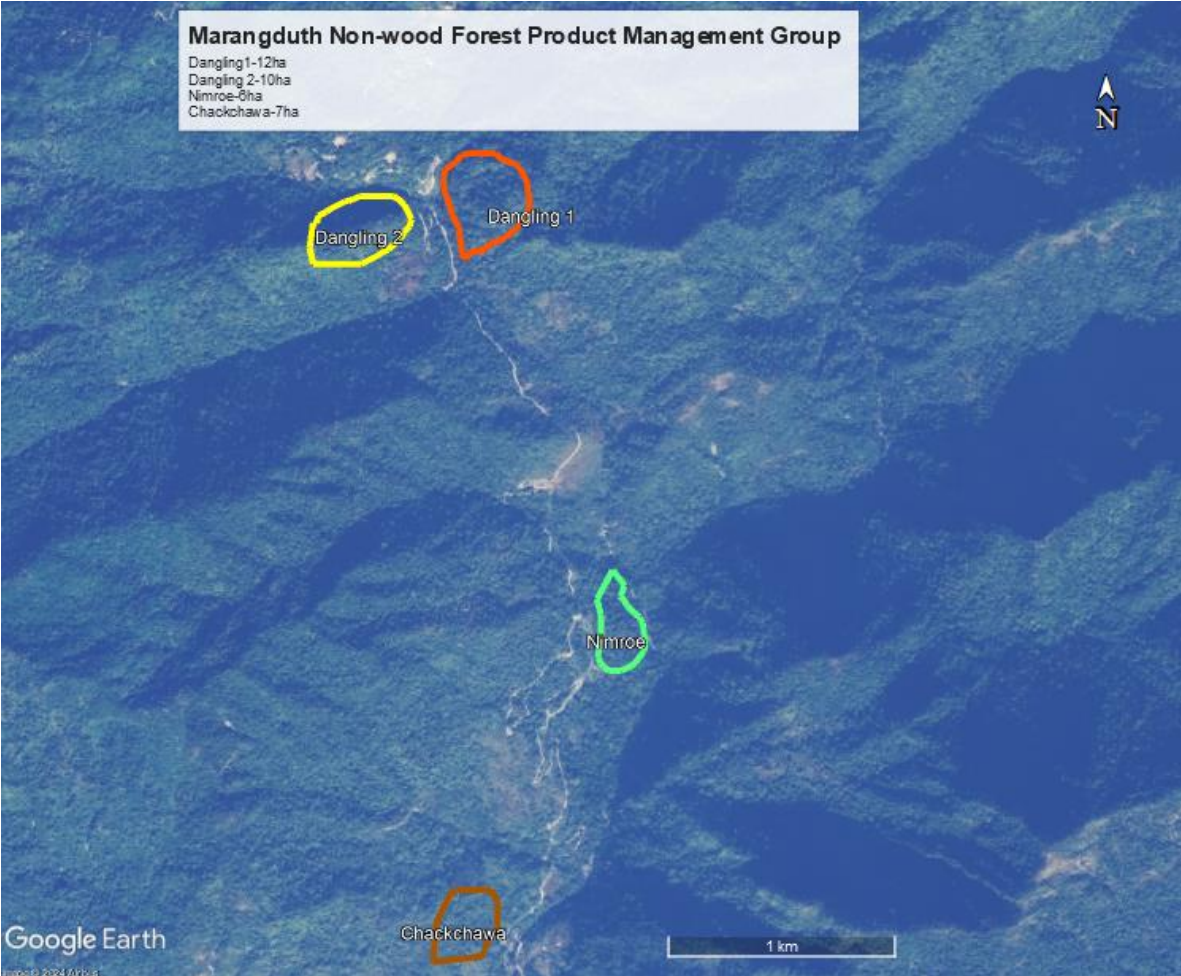


Figure 1. Map for NWFP management area

5. Marketing and sales plan

NWFPs collected by Marangduth NWFPs management & marketing group have planned to market & sell products to listed potential markets:

- 1. National Institute of Traditional Medicine Service (NITMS)
- 2. Food Cooperation of Bhutan (FCB) which will be operated as farmers shop at geog level
- 3. intruder who deals with the resources
- 4. Khengrig Namsum Cooperative at Tingtibi, Zhemgang
- 5. Local Markets at Panbang
- 6. Nearby border towns
- 7. Thimphu CSO market
- 8. Yula would be sold to Bjoka Tsharzo Gongphel Tshogpa and individual people of Bjoka.

The quantity, price of products and the value of expected sales are as listed in the table 3

Table 3. Estimated quantity of product and sales amount

NWFP	Potential Market	Estimated quantity for sale per season (unit)	Expected price per unit	Total (Nu)
<i>Neomicrocalamus andropogonifolius</i>	Bjoka Tsharzo Gongphel Tshogpa & Bjoka people individually	750 backloads (green)	400/bundle/mtr	187500
		750 backloads (dry)	850/bundle/mtr	262500
<i>Dendrocalamus hamiltonii</i>	Khengrig Namsum Cooperative at Tingtibi, Zhemgang	3500kgs (shoot)	12/kg	42000
		4000 pieces (stem)	20/piece	80000
<i>Diplazium esculentum</i> (fern)	Local Markets at Panbang	1500 bundles	30/=	45000
<i>Knema tenuinervia</i>	ITMS, border towns and intruder	3000kgs	80/=	240000
<i>Borinda grossa</i>	Border towns, intruder and Local Markets at Panbang	4500pieces	5/=	22500

<i>Piper betleoides</i>	Border towns, intruder and Local Markets at Panbang	2500 bundles	10/=	25000
<i>Oroxylum indicum</i>	Border towns, intruder and Local Markets at Panbang	2000pieces	20/=	40000
<i>Thysonolaenna latifolia</i>	Borderr towns, intruder and Local Markets at Panbang	2500 bundles	30/=	75000
<i>Musa</i>	Flower	2000pieces	20/piece	40000
	Stem	1500 stem	35/piece	52500
<i>Plectocomia himalayana</i>	Stem	2000mtr	10/mtr	20000
	Shoot	1000piece	30/piece	30000
<i>Tupistra nutans</i>	Flower	150kg	150/kg	22500
Total Amount projected				1184500

5.1 Marketing & Sales plan for prioritized bamboo products.

The community of Marangduth NWFP group has also projected the sales plan for prioritized *Dendrocalamus hamiltonii* resources mentioned table 5 below.

Table 4. Product from *Dendrocalamus hamiltonii* and expected product price

NWFP	Products (Local Name)	Unit	Rate per kg/Piece
<i>Dendrocalamus hamiltonii</i>	Boil Bamboo	Kg	25/=
	Bamboo pickle	Bottle 400gm	230/ bottle
	Bamboo pickle	Bottle 200gm	130/-
	Dry bamboo shoot	200gm	130/-pkt
	Preserve bamboo shoot	Kg	100/-kg
	Preserve bamboo dry shoot	Dray	300/-dray
	Tangloop	Piece	450/-piece
	Basket	Piece	800/-piece
	Bamboo hat	Piece	500/-piece
	Roong	Piece	1000/piece
	Zortay	Piece	100/-piece
	Shera	Piece	350/-piece

6. Forest Resource Assessment data for *Dendrocalamus hamiltonii*

Table 5. Resource assessment and data compilation form

Resource assessment compilation form for <i>Dendrocalamus hamiltonii</i>					
Block name:	Dangling 1, Dangling 2, Nimroe & Chackchawa	Dates of bamboo resource assessment:			
Sub-block (number)	Name of sub-block	Area (ha)	% of productive area	Productive area (ha)	Number of sample plots
1	Dangling 1	12	98	11.76	22
2	Dangling 2	10	96	9.6	16
3	Nimroe	6	95	5.7	12
4	Chackchawa	7	88	6.16	14

Age class	Number of culms	(Dangling1)	(Dangling2)	(Chackchawa)	(Nimroe)	Total in block
1+ year old bamboo culms	Total counted in sample plots	78	58	48	57	241
	Total harvestable in sample plots	68	52	42	55	217
	Total in block*	3120	2320	1920	2280	9640
	Total Harvestable in block*	2720	2080	2080	2200	8680
2+years old bamboo culms	Total counted in sample plots	72	50	45	58	225
	Total harvestable in sample plots	63	41	36	46	186
	Total in block*	2880	2000	1800	2320	9000
	Total Harvestable in block*	3150	1640	1440	1840	7440
3+years old bamboo culms	Total counted in sample plots	85	63	46	66	260
	Total harvestable in sample plots	70	54	39	55	218
	Total in block*	3400	2520	1840	2640	10400
	Total Harvestable in block*	2800	2160	1560	2200	8720
Total		18506	13038	10896	13817	55227
AHL		4359	3189	2515	3365	13431

AHL= 33% of the total stock of the 1+year old culms +2+year old culms+ 70% of 3+years old culms

In which the 33% of the stock of the 1+year old and the 2+years old culms should not exceed the total of the harvestable culms of the combined age classes.

* As calculated per productive area in sub-block/block (calculate the average per sample plot and divide by 0,025(10*25)/10000) to get the number of culms

Per ha, multiply with the productive area (ha) of sub--block/block to get the total of culms per sub-block/block: compare the result with the culms considered harvestable by the villagers (calculated in the same way).

6.1 Demand & annual harvesting limit for prioritized Bamboo products

Table below denotes the demand of community members for *Dendrocalamus hamiltonii* shoot and culms and its annual harvesting limits.

Table 6. Annual demand and AHL

Total Annual Demand	Annual Harvesting Limit	Variation	Remarks
4000 pieces culms	13431 pieces of culms	9431 pieces of culms	Surplus

7. Marketing outcome in the previous plan.

During the plan review, the cost analysis for sale forecast and actual income generated from the sale of different NWFPs products were compared and computed as mentioned below. The community couldn't implement the activities as per the previous plan due to the Covid-19 pandemic. However, they could manage to harvest bamboo shoot as per the market demand and earned sum of Nu.40,000.00

Table 7. Product harvested and income generated in previous plan period

NWFPs	Estimated qty. for sale	Sold qty.	Estimated price	Total Nu.	Sold price	Income generated
<i>Neomicrocalamus andropogonifolius</i>	750 backloads (green)		250/=			Nil
	750 backloads (dry)		350/=			Nil
<i>Dendrocalamus hamiltonii</i>	3500kgs (shoot)		12/=			40000/-
<i>Diplazium esculentum</i> (fern)	1500 bundles		30/=			Nil
<i>Knema tenuinervia</i>	3000kgs		80/=			Nil
<i>Borinda grossa</i>	4500pieces		5/=			Nil
<i>Piper betleoides</i>	2500 bundles		10/=			Nil
<i>Oroxylum indicum</i>	2000pieces		20/=			Nil
<i>Thysonolaenna latifolia</i>	2500 bundles		30/=			Nil

8. Marketing Arrangement

Following were the marketing strategies planned for the Marangduth Non—wood Forest Management Group during the group formulation.

1. Products will be collected & transported up to the road point by the members
2. Products like cane shoot, mushroom, damroo, fern etc will be sold in local market
3. Piper will be sold to nearby Indian town and other intruders
4. Yula to local markets & finished products at Thimphu
5. Cane to Bjoka Tsharzo Gongphel Tshogpa and other traders.
6. *Dendrocalmus himaltonii* to KNC, Thimphu and other traders.

8.1 Procedures for NWFP harvesting, marketing and sharing of income

Activity	Brief description of how the activities is carried out
Harvesting of products	• The Chairperson of NWFP group will obtain harvesting/collection permit from forest office in writing. • The Chairperson will announce starting date for harvesting only after obtaining required permits from forest office. • Only one member from each household will be collect NWFP for the first week and after that no limits is set for member involvement for collection. • Harvesting prescriptions will be strictly followed by the collectors. • Chairperson shall inform Range Office of completion of harvest.
Postharvest practices	• The collectors are responsible for post-processing of NWFPs.

Marketing of products	• The NWFPs will be handed over to the executive members for marketing after post-processing. • The executive members will sell the produce in attendance of the group members to maintain transparency. In future the group plans to construct a common sale counter at the center to facilitate marketing of the NWFPs products.
Identification of buyers	• The Chairperson supported by other executive members will monitor and explore the demand from nearby institutes, agents, companies and other business groups. • Support will also be sought from Department of Agriculture and Marketing Cooperatives (DAMC)
Price fixing	• The executive members will coordinate.
Income sharing	• . The income generated from the sale of NWFPs will directly share among the members. All the members will have equal right for sustainable harvesting of products as per the entitled lists in the plan. The group will also utilize the common fund during the community rituals and natural calamities with basic understanding among the members
Raising of community development fund	• A joint account in the name of the group will be opened under BDBL Branch Office located at Panbang. The account shall be managed by the committee members on behalf of the group. • Annual membership fees, penalties and certain amount of sale proceeds will be deposited in the community development fund.
Payment to committee members for their service	• 5% of total income from the sale of products will be deposited into group's account & 15% from the sale will be shared within the committee workers as work charges.

9. Benefit to the members

The income generated from the sale of NWFPs will directly share among the members. All the members will have equal right for sustainable harvesting of products as per the entitled lists in the plan. The group will also utilize the common fund during the community rituals and natural calamities with basic understanding among the members.

9.1 Financial projection for prioritized species and benefits to the members from NWFP sales

Table 8. Financial projection

Sl. No	Particulars	Products (names)				Total Amount (Nu)
		Dendrocalamus hamiltonii		Tupistra nutans	Thysolaenna latifolia	
		Shoot	Culm			
1	Revenues					
A	Total quantity (unit)	3500kg	4000pieces	150kg	2500bundles	
B	Market price (Nu per unit)	12	20	150	30	
C	Total revenues from forest product sales (a*b)	42000	80000	22500	75000	219500
A	Total revenues					219500
2	Expenses					
A	Packaging materials			-	-	-
B	Royalty	4000 pieces@12			-	48000
C	Transportation costs					15000

D	Compensation fee to executive committee for marketing products	-	-	-	-
E	Contribution to community development fund	5% of total revenue	-	-	10975
F	Incidental expenses	-	-	-	-
B	Total expenses				73975
C	Total benefit (net-income) of group members (Total revenues A – Total expenses B)				145525
D	Average income to group members (C/ no of group members)				5821
E	Price per unit paid to group members (C/1a)				14.34

Table 9. Financial projection to CDF

Sl. No	Particulars	Amount (Nu)
	Revenues	
1	Membership registration fees (25hh @100)	2500
2	5% collection from total revenue	10975
A	Total revenues	13475
	Expenses (only include expenses that will be paid from the group funds)	-
		
B	Total Expenses	-
C	Balance: Total revenues – Total expenses	13475

Part B: Group Regulation and Norms (By-Law)

NWFP management & marketing group	
Membership arrangement and resignation	Any household with census recorded in Marangduth village and residing in the area can join the group. However, acceptance in the group will be made only by the group members through assembly & submitting letter of interest to the committee members.
Responsibilities and rights of members:	<u>Responsibilities and rights of members:</u> • Implement annual activities that are reflected in the management plan. • Manage the area in adherence to the management plan and refrain from doing illegal activities. • Attend ad-hoc/ planned meetings or activities for development of the group. • Attend any training related to NWFP. • Choose or reject committee members for better management of the group. <u>Procedure for becoming new members:</u> • Any household wishing to become members in the group shall submit application to the Chairperson. • The applicant should be 18 years of age to become a member. • The application shall be put forward in the general meeting for discussion and every member has right to accept or reject the application. • Membership fee of Nu.1500/- shall be levied upon registration. <u>Procedure for leaving the group:</u> • Any member wishing to discontinue their membership has right to do so. However, he/she should submit application to the Chairperson. • He/she can leave the group after discussing with the committee members. • Member leaving the group will not be allowed to avail their benefits from the group fund. • If their children wish to join as a member in future, they have to follow the procedure set to become new member.
Executive committee of NWFP management and marketing group	
Chairperson	Name of Chairperson: Dorji Wangdi Contact No.17680976 <u>Responsibilities of Chairperson:</u> • Chair the group meetings in which the group will have half yearly meeting twice in a year & a general meeting • Explore market for NWFPs. • Obtain collection permit from forest office during the onset of harvesting season. • Coordinate and plan activities for the user group along with the members and other stakeholders. • Coordinate and conduct meetings with members for development of the group. • Resolve conflict among group members. • Submit annual progress report to forest office. • Maintain record of all the group activities.

Secretary	Name of Secretary: Kinzang Thinley Contact No. <u>Responsibilities of Secretary:</u> • Officiate in absence of Chairperson. • Assist Chairperson in coordinating and planning group activities. • Explore market for NWFPs. • Maintain records of any activities conducted by the group. • Maintain attendance record while conducting meeting and group activities. • Record minutes of meeting.
Treasurer	Name of Treasurer: Tandin Wangchuk & Tashi Zangmo Contact No. 17683694 <u>Responsibilities of Treasurer:</u> • Officiate in absence of Chairperson and Secretary. • Collect annual membership fees, fines, amount from sale proceeds and deposit in the group account. • Maintain proper financial records in favor of the group. • Present financial status/ progress report during general meeting. • Assist Chairperson and Secretary in coordinating and planning group activities.
Tenure of committee member	• Their tenure shall be till the plan period ends. • However, if a committee member does not fulfill his/her responsibilities, misbehave and threaten the group's network, he/she shall be immediately replaced and be deprived of such opportunity in future.
Incentives to the committee members	• If the committee members have worked positively towards betterment of group, he/she shall be re-elected. This is to reward the individual for his/her dedicated service to the group.
Observer	Name: Dorji Responsibilities of Observer • Act as an Audit • Observation of overall work
Resoop	• Conduct Patrolling twice in a year • Report illegal activities to the committee members
Sources for generation of group fund	
Membership registration fees. Annual contribution Collection from sale proceeds	• Membership fees of Nu.100 per member household shall be collected annually. • The amount deposited will be used as pre-financing cost for marketing of NWFPs products • Amount realized as penalties shall be deposited in the group fund. • 5% of total products sold will be deposited to group account
Group fund management and utilization	• A joint account in favor of the group shall be opened under BDBL, Panbang. • Annual membership fees, fines and amount from sale proceeds shall be deposit in the group account. • The joint account will be managed by Executive Committee members. • Any expenditure incurred in the name of the group will be met from community development fund. • When the fund accumulates more than two hundred thousand, the fund will be utilized for community rituals, supporting the poor and disaster affected household. • If the group gets dissolved under unforeseen circumstances, the entire fund accumulated shall be equally divided among the members.

	Welfare scheme will be decided by community member upon open discussion with the member
Conflict resolution	- Disputes between the members will be resolve by committee if not, - The conflicts that arise among the group members about their group activities & certain areas of NWFP harvesting will be put forward to Geog Tshogdu (GT) followed by Dzongkhag Tshogdu (DT) - If the conflict arises between NWFP group and outsider, case will be forwarded to DFO and Geog Administration by the committee
Meeting and purpose	In order to plan and discuss activities for the management of NWFPs, meeting will be conducted as follows; General Assembly: All the members shall attend the meeting to discuss and plan activities before harvesting season; during April month every year. Committee member: This meeting is mainly to discuss the planned activities as well as prepare progress reports (financial and physical) during the November month.
Penalties	- If the group members' violet the harvesting rules for NWFPs, he/she should pay the fine of Nu. 250/= and will not allow to harvest the products in next harvesting season - Any group members selling the products to outsider without notice of group will penalize fine of Nu. 1500/= - If outsider collects the products from the group's demarcated areas, they will be fined of Nu. 500/= if their intention is to commercialize for income generation. - If the collection area is being damaged due to cattle grazing, the owner will be fined Nu.150/= per cattle - Member inactive after one year will be excluded from the group. - Income generated in group will be shared with the active member except inactive member will not be entitled.

Powers of the Department

The Department has the right to terminate the NWFP management and marketing group in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.

The deregistration, cancellation, termination and reinstatement of the NWFP plan shall be guided by Section 261 to 265 of the Forest and Nature Conservation Rules and Regulations, 2023.

Appendix I: Overview of *Dendrocalamus hamiltonii* production area

Block name	Sub-block	Total area (Ha)	Production area (%)	Production area (Ha)
Dangling 1	Dangling 1	12	98	11.76
Dangling 2	Dangling 2	10	96	9.6
Chackchawa	Chackchawa	6	95	5.7
Nimroe	Nimroe	7	88	6.16
				33.22

Appendix II: Block description for *Dendrocalamus Hamiltonii* collection

Dangling 1 Block

Block description form for <i>Dendrocalamus hamiltonii</i>							
Date:	16-01-24						
Village:	Dongdhor		Name of bamboo area:Dangling1				
Block name:	Dangling 1		Total area (ha)		12		
Access	What is the walking distance from the nearest village to the block ?						
	Less than 3 hours		Within a day		More than a day		
	What is the walking distance from the block to the nearest road ?						
	Less than 3 hours		Within a day		More than a day		
Slope	Very steep >100%	Steep 70-100%		Moderate 45-70%	Gentle <45%		
Forest Description	Dominant and associate species, canopy cover:						
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots		
	1		23.33	98	16		
	2						
	3						
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None			
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable			
	1. Dangling	Yes					
	2						
	3						
Current management	Current management practices in entire block:						
Current threats	Threats	Description					
	Fire						
	Grazing						
	Illegal exploitation	Illegal collection by commuters since the area is farflung from the community.					
	Over-exploitation						
	Other:						
Opportunities	Future opportunities for the entire block: High production area if managed sustainably						

Dangling 2 Block

Block description form for Dendrocalamus hamiltonii						
Date:	18-01-24					
Village:	Dongdhor		Name of bamboo area:Dangling 2			
Block name:	Dangling 2		Total area (h10			
Access	What is the walking distance from the nearest village to the block ?					
	Less than 3 hours		Within a day		More than a day	
	What is the walking distance from the block to the nearest road ?					
	Less than 3 hours		Within a day		More than a day	
Slope	Very steep >100%		Steep 70-100%		Moderate 45-70%	Gentle <45%
Forest Description	Dominant and associate species, canopy cover:					
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots	
	1		24.71	16	14	
	2					
	3					
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None		
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable		
	1. Dangling 2	Yes				
	2					
	3					
Current management	Current management practices in entire block: NWFP managed area					
Current threats	Threats	Description				
	Fire					
	Grazing					
	Illegal exploitation	Illegal collection by commuters since the area is farlung from the community.				
	Over-exploitation					
	Other:					
Opportunities	Future opportunities for the entire block: High production area if managed sustainably					

Nimroe Block

Block description form for Dendrocalamus hamiltonii										
Date: 20-01-24										
Village: Nimroe		Name of bamboo area:Nimroe								
Block name: Nimroe		Total area (ha) 6								
Access	What is the walking distance from the nearest village to the block ?									
	Less than 3 hours			Within a day		More than a day				
	What is the walking distance from the block to the nearest road ?									
	Less than 3 hours			Within a day		More than a day				
Slope						Very steep >100%	Steep 70-100%		Moderate 45-70%	Gentle <45%
Forest Description						Dominant and associate species, canopy cover: Daubanga grandedolia, Schima wallachi, Gmelia arborea, Calicarpa arborea, Terminalia mycrocarpa				
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots					
	1		14.8	5.7	8					
	2									
	3									
Bamboo flower	Sporadic	Gregarious	Last time of mass flowering	None						
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable						
	1. Nimroe	Yes								
	2									
	3									
Current management	Current management practices in entire block: NWFP managed area									
Current threats	Threats	Description								
	Fire									
	Grazing									
	Illegal exploitation	Illegal collection by commuters since the area is farflung from the community.								
	Over-exploitation									
	Other:									
Opportunities	Future opportunities for the entire block: High production area if managed sustainably									

Chackchawa Block

Block description form for Dendrocalamus hamiltonii								
Date:	22-01-24							
Village:	Chackchawa			Name of bamboo area:Chackchawa				
Block name:	Chackchawa			Total area (ha)		7		
Access	What is the walking distance from the nearest village to the block ?							
	Less than 3 hours			Within a day		More than a day		
	What is the walking distance from the block to the nearest road ?							
	Less than 3 hours			Within a day		More than a day		
Slope	Very steep >100%	Steep 70-100%		Moderate 45-70%		Gentle <45%		
Forest Description	Dominant and associate species, canopy cover: Daubanga grandedolia, Schima wallachi, Gmelia							
Sub-block	No.	Density	Area (acres)	Production area (%)	No. of sample plots			
	1		17.3	6.16		8		
	2							
	3							
Bamboo flowering	Sporadic	Gregarious	Last time of mass flowering	None				
Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable				
	1. Chackchawa							
	2							
	3							
Current management	Current management practices in entire block: NWFP managed area							
Current threats	Threats	Description						
	Fire							
	Grazing							
	Illegal exploitatio	Illegal collection by commuters since the area is farflung from the community.						
	Over-exploitation							
	Other:							
Opportunities	Future opportunities for the entire block: High production area if managed sustainably							

Appendix III: Resource assessment form for Dendrocalamus hamiltonii

Dangling 1 Block

Date: 16/1/2024

Block: Dangling 1

Sample plot no: 1

Aspect: NE

Village: Dongdhor

Local name of resource assessment area: Dangling 1

Slope: 59%

Altitude: 1046m

Recorder: Sherab Dorji & Ugyen Dorji

Slope correction: 1.161

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	5	5	7	6	3	4	4	11
2	Good	6	7	6	5	7	5	5	17
3	Congested	15	10	8	11	9	8	10	27
4	Congested	11	11	9	12	11	8	9	28
5	Congested	9	12	10	17	10	8	15	33
6	Congested	12	15	14	11	13	12	9	34
7	Good	6	6	4	7	5	4	6	15
8	Good	5	5	7	7	4	6	4	14
9	Congested	12	10	13	13	8	11	12	31
10	Damaged	2	2	1	2	1	1	0	2
		83	83	79	91	71	67	83	212

Dangling 2 Block

Date: 18/1/2024

Block: Dangling 2

Sample plot no: 2

Aspect: NW

Village: Dongdhor

Local name of resource assessment area: Dangling 2

Slope: Less than 45%

Altitude: 1025m

Recorder: Kiran Mongar & Sha Kinley Doya

Slope correction: 1.093

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	7	6	7	9	6	5	8	19
2	Good	7	4	3	6	4	2	5	11
3	Damaged	1	2	1	2	1	1	1	3
4	Congested	12	15	13	10	13	11	9	33
5	Good	3	4	5	7	4	4	5	13
6	Good	5	6	4	7	4	3	7	14
7	Damaged	1	3	0	2	3	0	2	5
8	Congested	12	14	13	12	13	12	11	36
9	Good	5	4	4	8	4	3	6	13
		53	58	50	63	52	41	54	147

Chackchawa Block

Date: 22/1/2024

Block: Chackchawa

Sample plot no: 3

Aspect: NW

Village: Marangduth

Local name of resource assessment area: Chachchawa

Slope: 54%

Altitude: 541m

Recorder: Sherab Dorji & Ugyen Dorji

Slope correction: 1.136

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Good	6	6	4	7	6	2	6	14
2	Good	7	7	5	2	6	5	2	13
3	Congested	11	12	11	9	9	10	7	26
4	Good	6	4	4	7	4	3	5	12
5	Good	4	3	4	5	3	3	4	13
6	Congested	9	11	10	9	9	9	8	26
7	Good	3	5	7	7	5	4	7	16
		46	48	45	46	42	36	39	94

Nimroe Block

Date: 20/1/2024

Block: Nimroe

Sample plot no: 4

Aspect: SW

Village: Marangduth

Local name of resource assessment area: Nimroe

Slope: Less than 45%

Altitude: 524m

Recorder: Kiran Mongar & Sha Kinley Doya

Slope correction: 1.093

No of clump	Clump condition*	Shoots/less than 1 year old culm	Bamboo stock			Harvestable culms **			Total no of culms
			1+ year (no)	2+years (no)	3+year (no)	1+ year (no)	2+years (no)	3+year (no)	
1	Congested	8	9	9	12	8	7	11	26
2	Good	3	7	5	7	7	4	5	16
3	Good	5	5	8	4	4	5	4	13
4	Good	2	6	5	8	6	4	6	16
5	Congested	8	10	10	15	8	10	12	30
6	Good	7	8	6	4	7	5	4	16
7	Good	4	6	6	8	4	3	2	9
8	Congested	9	12	9	8	11	8	6	25
		46	58	58	66	55	46	50	151

Appendix IV: Annual NWFP reporting form

Dzongkhag/Dungkhag:
Name of NWFP management and marketing group:
Gewog:
Villages:
Total number of Households:

NWFP Species	Quantity collected in a year	Qty sold in a year	Price received by the group per unit	Total income (Nu)	Description on resource availability	Remarks

Appendix V: List of NWFP Group Members

Sl. No.	Name	CID No.	Thram No.	House No.	Village
1	Dorji Wangdi	12005000826	695	Wa-5-182	Marangduth
2	Dawa	12005000606	137	Wa-5-37	Mucheni
3	Rinchen Jamtsho	12005000976	88	Wa-5-179	Marangduth
4	Thongjay	12005000885	95	Wa-5-187	Marangduth
5	Jatsho	12005000813	445	Wa-5-19 Nil	Marangduth
6	Karma	12005003602	86	Wa-5-285	Prikchang
7	Tsheddar	12005000824	91	Wa-5-295	Marangduth
8	Pema	12005001364	212	Wa-5-129	Marangduth
9	Dorji	12005003444	314	Wa-5-320	Marangduth
10	Rinchen Penjor	12005003070	991	Wa-5-240	Prikchang
11	Tshering	12005000934	552	Wa-5-191	Marangduth
12	Choki	12005002869	93	Wa-5-193	Marangduth
13	Ngada	12005000771	305	Wa-5-178	Marangduth
14	Pelzang	12005001422	410	Wa-5-260	Dongdhor
15	Pema Thinley	12005000858	97	Wa-5-184	Marangduth
16	Kencho	12005000815	92	Wa-5-181	Marangduth
17	Sangay	12005000362	249	Wa-5-237	Doduphu
18	Rinchen	12005003697	516	Wa-5-242	Prikchang
19	Sangay Dorji	12005000845	98	Wa-5-183	Marangduth
20	Kinzang Thinley	12005000800	272	Wa-5-180	Marangduth
21	Thaupachen	12005000926	100	Wa-5-189	Marangduth
22	Gembo Dorji	12005000754	35	Wa-5-177	Marangduth
23	Rinchen Jampel	12005000867	99	Wa-5-185	Marangduth
24	Jigme Dorji	12005003611	171	Wa-5-236	Prikchang
25	Pema Lhamo	12005003707	43	Wa-5-243	Duolophu
26	Namgyel	12005000935	551	Wa-5-Nil	Marangduth
27	Lungten	12005000999	581	Wa-5-345	Dongdhor
28	Leki Duba	12005002588	804	Wa-5-Nil 27	Dongdhor